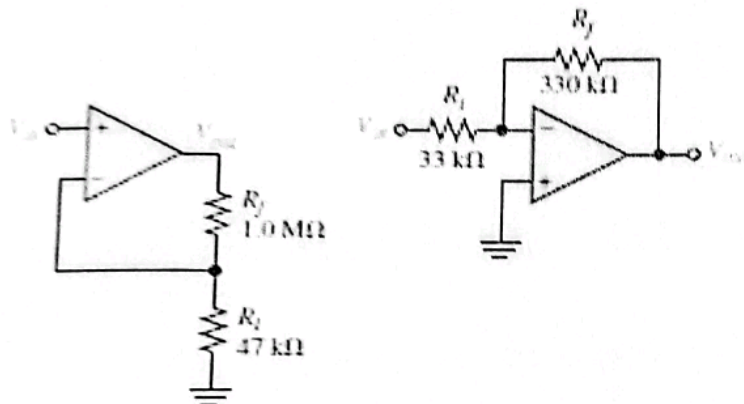
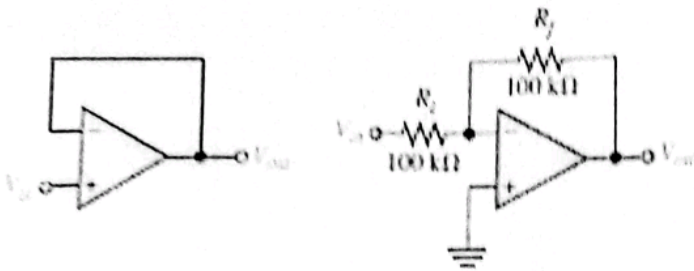
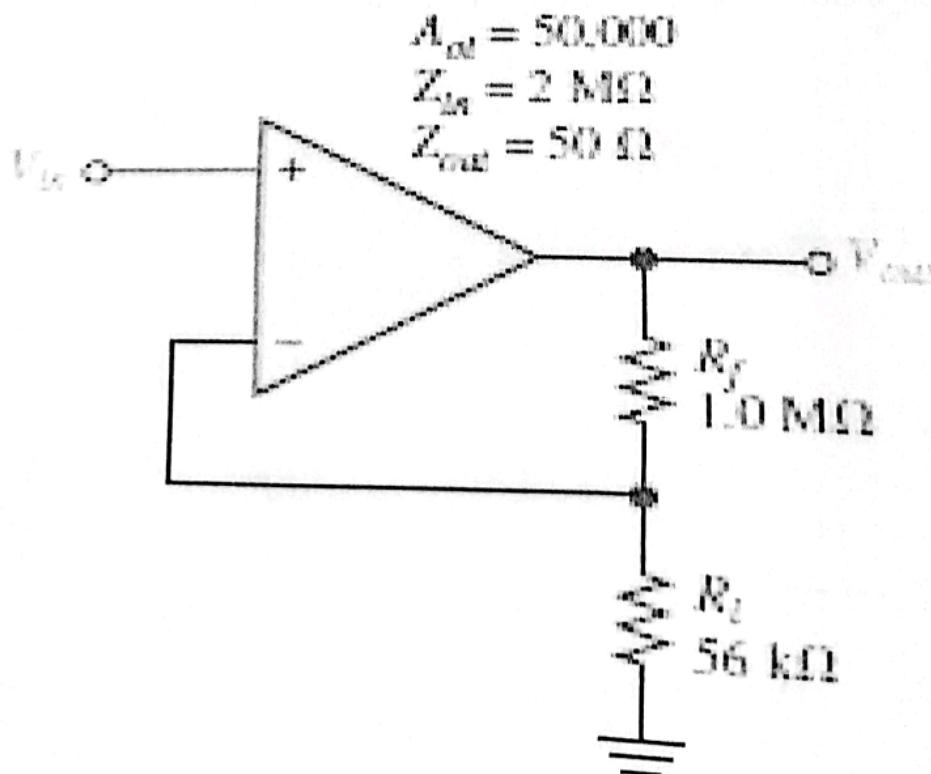


TUTORIAL

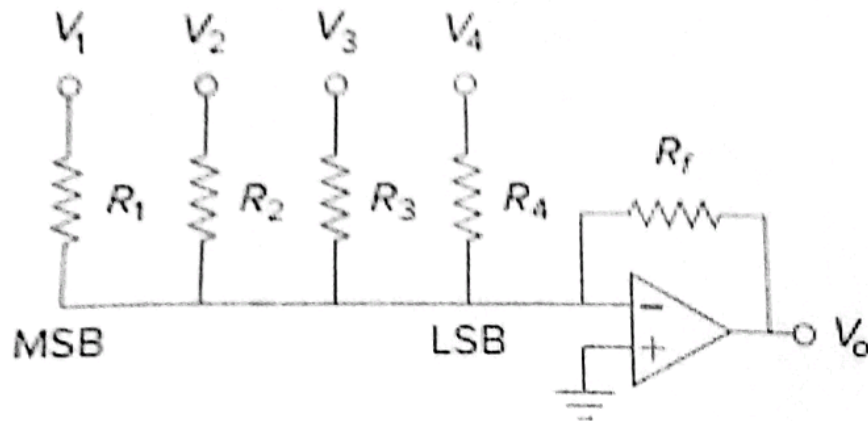
1. Compare a practical op-amp to an ideal op-amp.
2. Find the gain of each amplifier below:



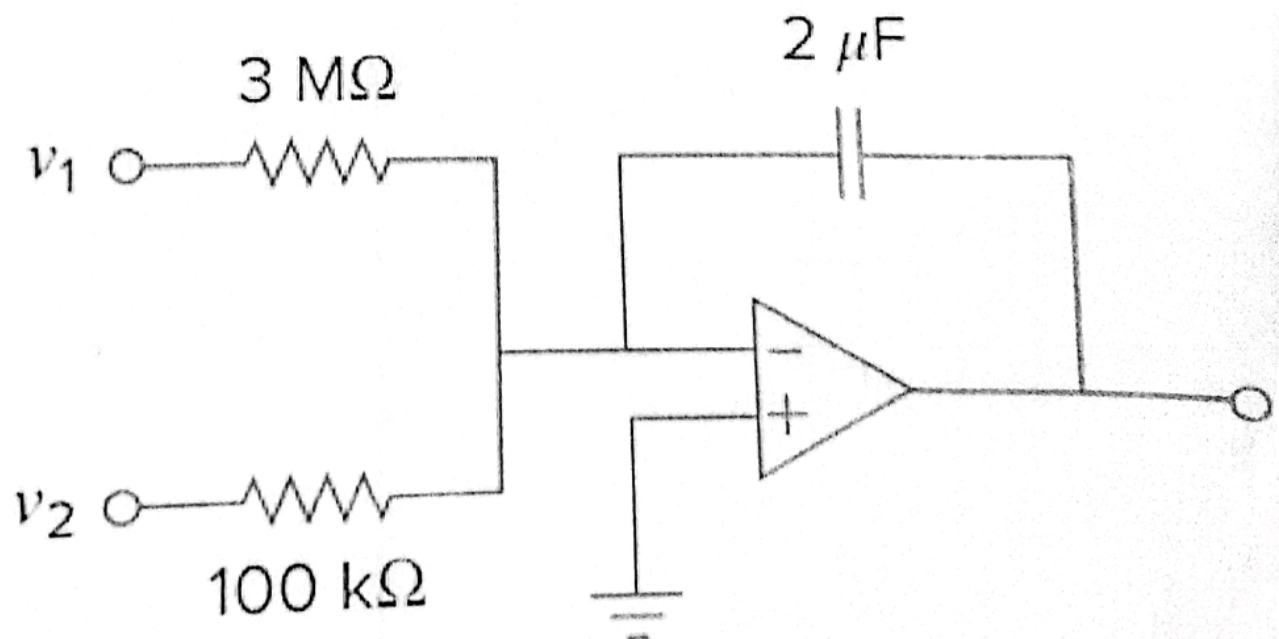
3. Determine the input and output impedances for the amplifier configuration below:



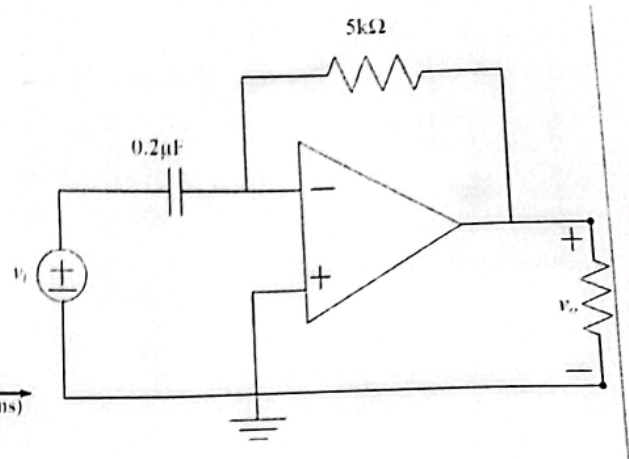
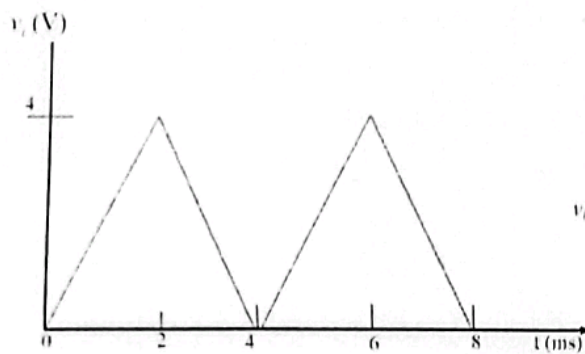
4. In the DAC circuit below, $R_f = 10 \text{ k}\Omega$, $R_1 = 10 \text{ k}\Omega$, $R_2 = 20 \text{ k}\Omega$, $R_3 = 40 \text{ k}\Omega$, and $R_4 = 80 \text{ k}\Omega$. Obtain the analog output voltage for the digital inputs [0000], [0001], [0010], ..., [1111]. Consider $V_{\text{ref}} = 1 \text{ V}$.



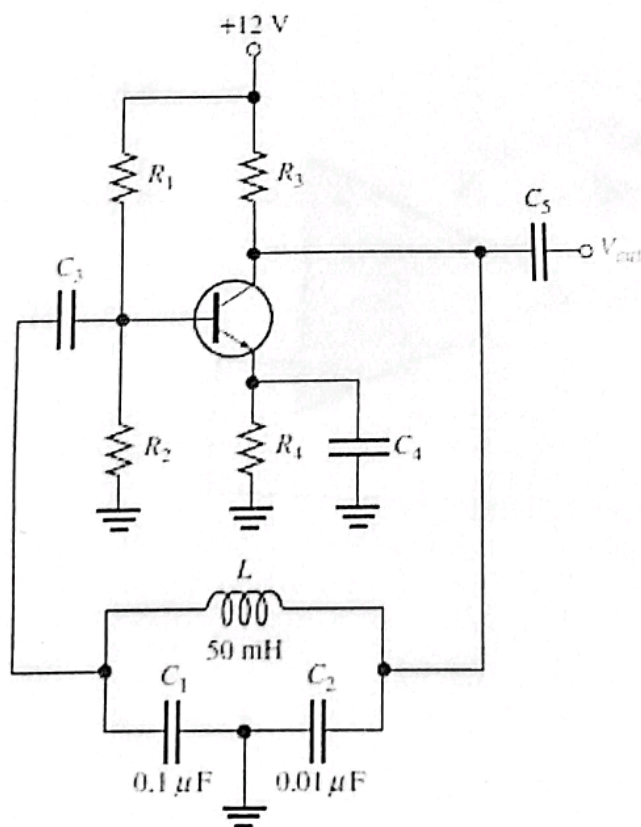
5. If $V_1(t) = 10\cos(2t) \text{ mV}$ and $V_2(t) = 0.5t \text{ mV}$ below, Find $V_o(t)$ for $t > 0$. Assume that the voltage across the capacitor is zero at $t = 0$.



6. Sketch the output voltage waveform $V_o(t)$ for the circuit shown, given the input voltage waveform below.



7. What is the basic difference between the Colpitts and the Hartley oscillators?
8. Determine the frequency for the oscillator in figure below. Assume there is negligible loading on the feedback circuit and that its Q is greater than 10.



9. (a) Determine the value of R_f necessary for the circuit below to operate as an oscillator.

(b) Determine the frequency of oscillation.

